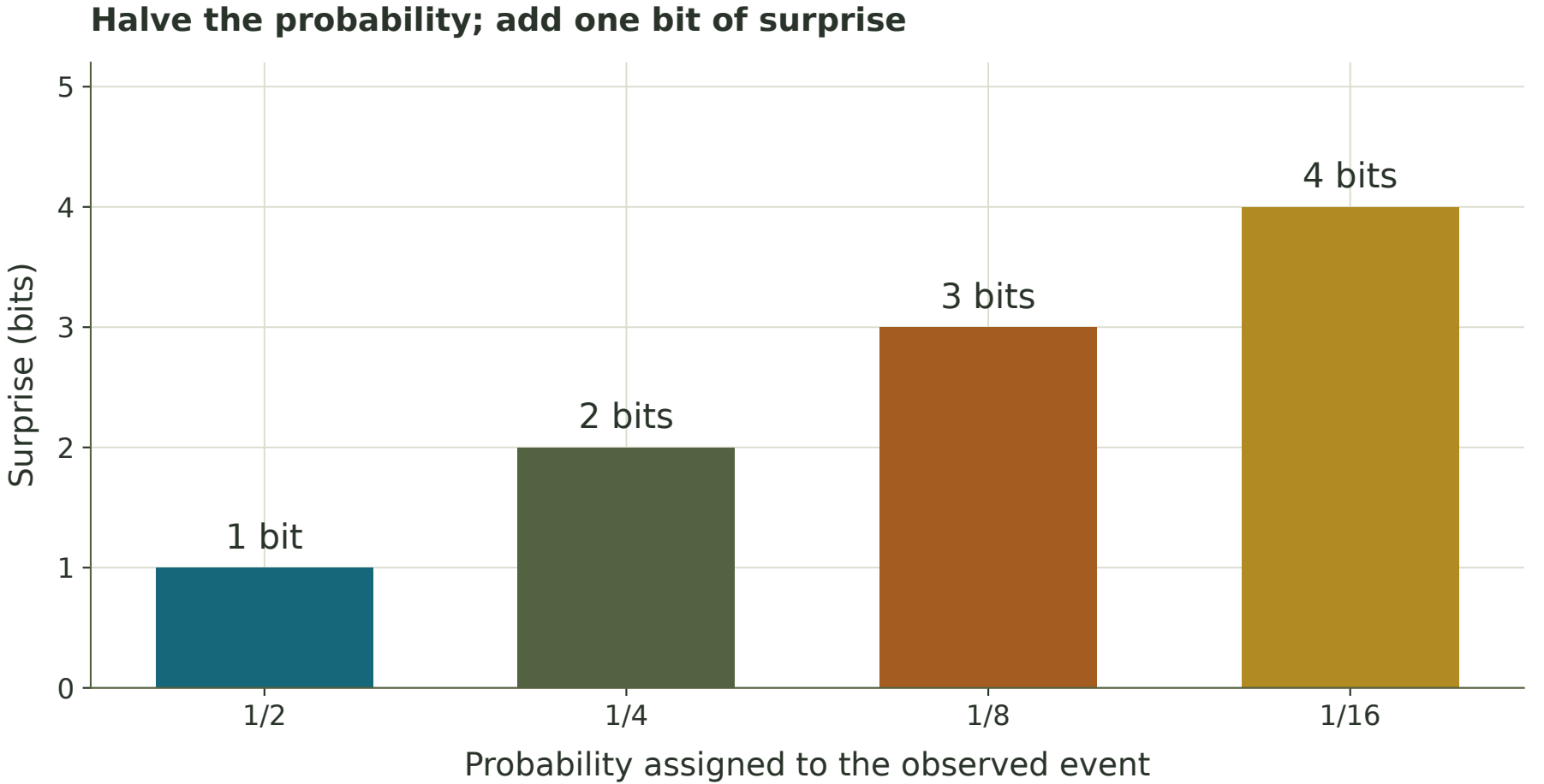
Halve the probability; add one bit of surprise
Surprise (bits)
0
1
2
3
4
5
1 bit
2 bits
3 bits
4 bits
1/2
1/4
1/8
1/16
Probability assigned to the observed event